

# Watermarking: Applications and Current State of the Art

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# Outline

- Introduction
- Applications
- Requirements
- Data Hiding Approaches
- Future Directions and Conclusion

# Data Embedding

## ■ Content description & management

- IPR data: originator, creation parameters, version control, editing

## ■ Rights management and protection

- audit trail, monitoring, copy/no copy control, linking to a player

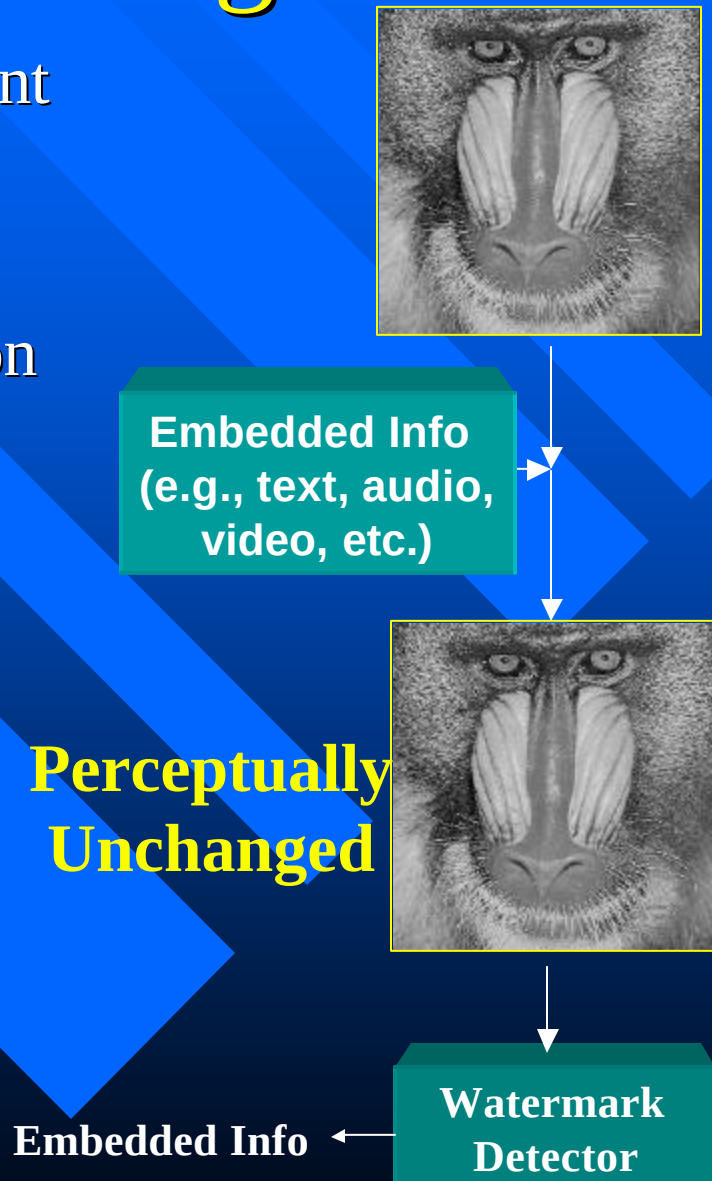
## ■ Fraud detection

## ■ E-commerce

- hyperlink to sales site

## ■ Customized/Enhanced media delivery

- customized commercials
- customized viewing



# Application

## Rights Management and Protection

### ***SDMI World***

- *Compliant Devices/Players*
- *Secure Media*
- *Rippers*
- *DRM rules- Copy/No Copy*

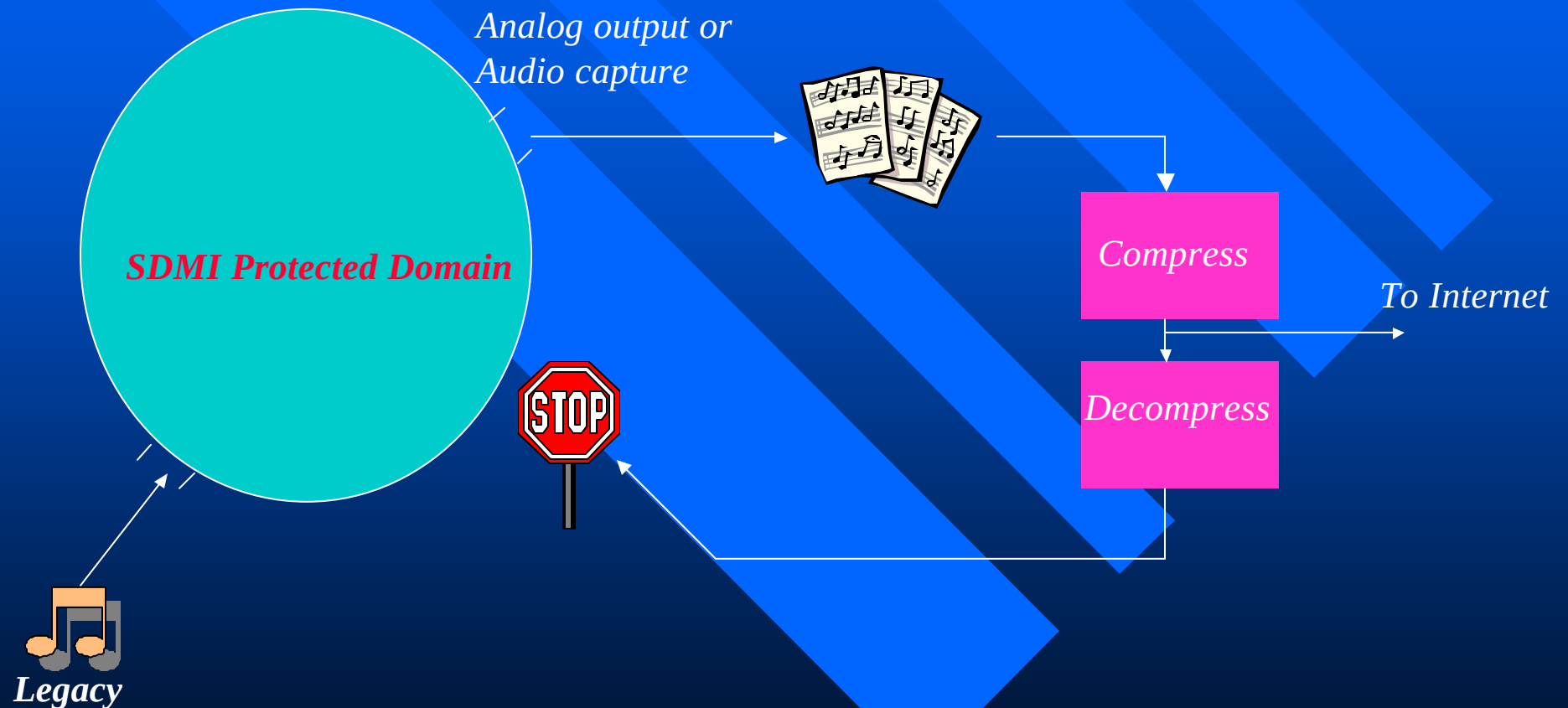
### *SDMI attempts to:*

- Create an environment for legitimate distribution of music
- Eliminate illegal download and swapping of copyrighted music

*SDMI: Secure Digital Music Initiative – Consortium of 200+ technology companies*

# Application

## Rights Management and Protection



# Application

## **Rights Management and Protection**

- Robust Watermark
  - Copy Control/Source Information
  - Fragile watermark assertion
- Fragile Watermark
  - Disappearance indicates compression

# SDMI Requirements

## Robust Watermark

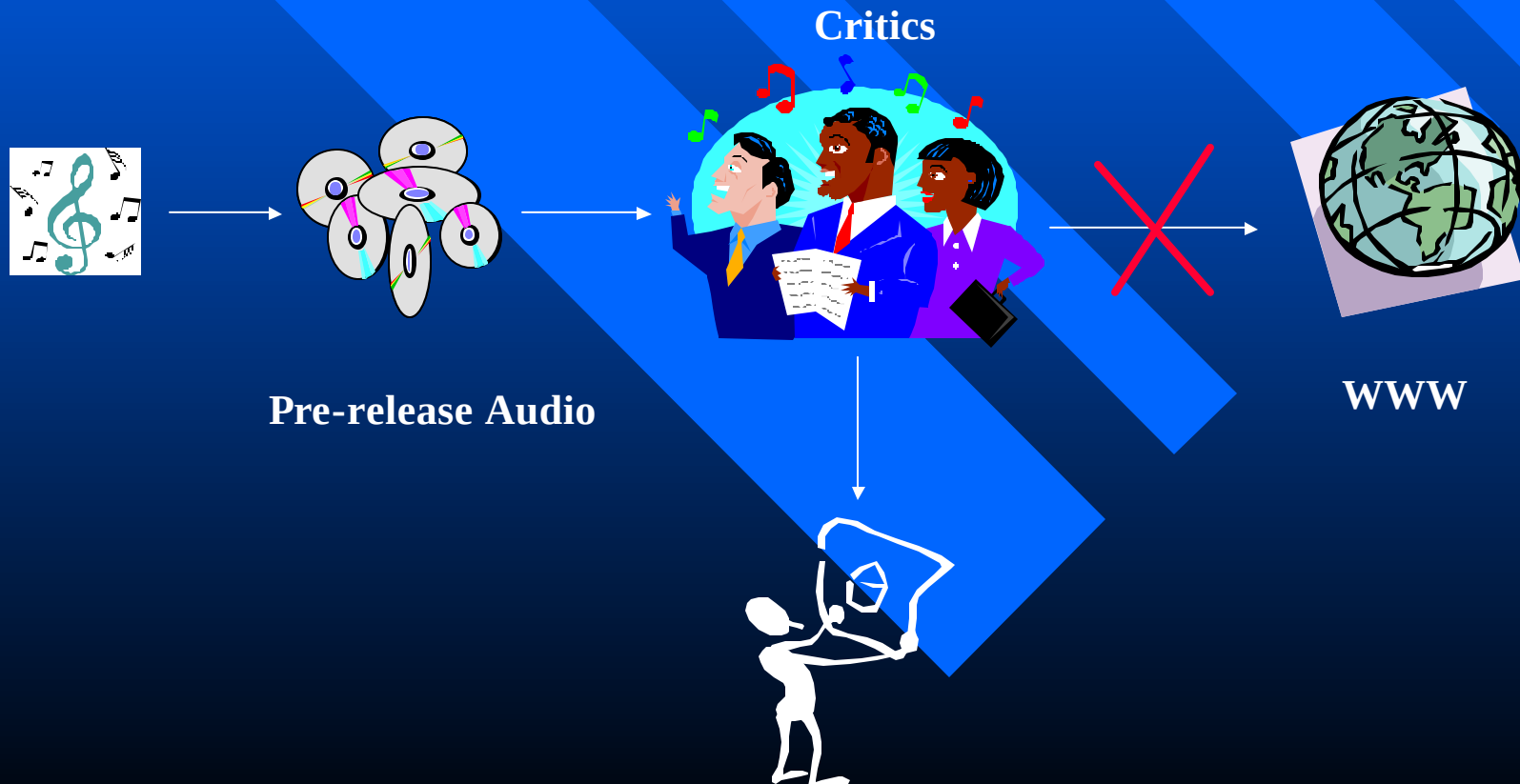
- Transparency
- Two successive D/A and A/D conversions
- Linear Speed Change of +/-10 %
- Audio coding (MPEG, AC-3, AAC, ...)
- Dynamic range reduction
- Additive noise
- Down-mixing and Surround sound processing
- Echo Addition
- Bandpass filtering
- Low complexity

## Fragile Watermark

- Transparency
- Anything but coding
- Low complexity

# Application

## ■ Watermarking of Pre-release Audio

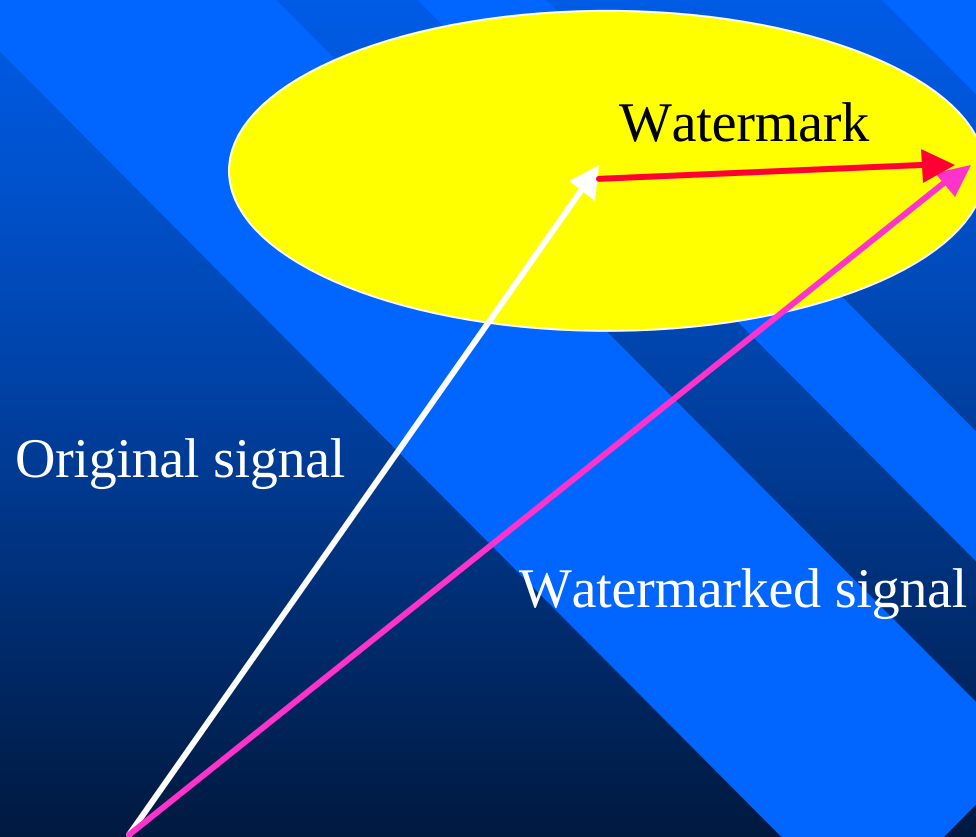


# Data Embedding Issues

- Transparency
- Capacity
- Robustness
- Security

# Masking and Data Embedding

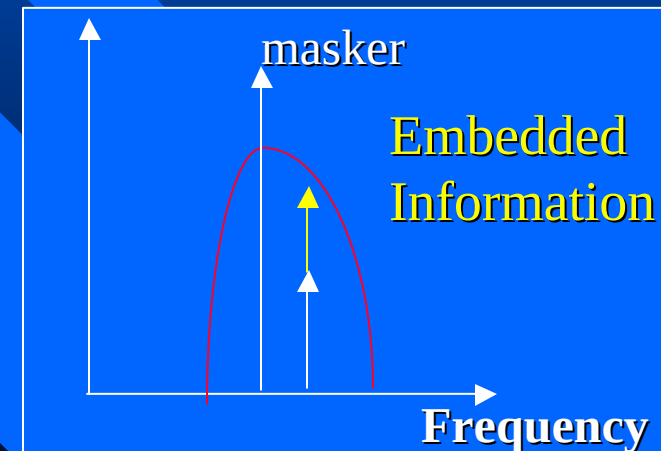
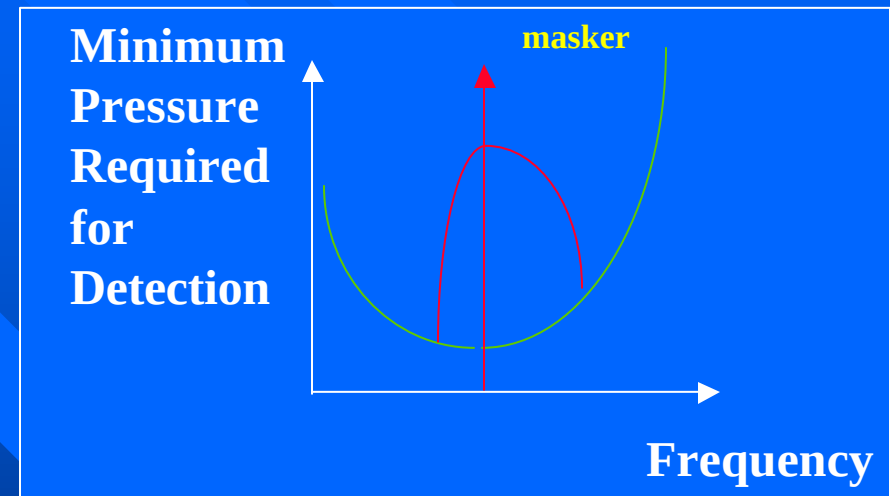
Add watermark such that watermarked signal is perceptually equivalent to original



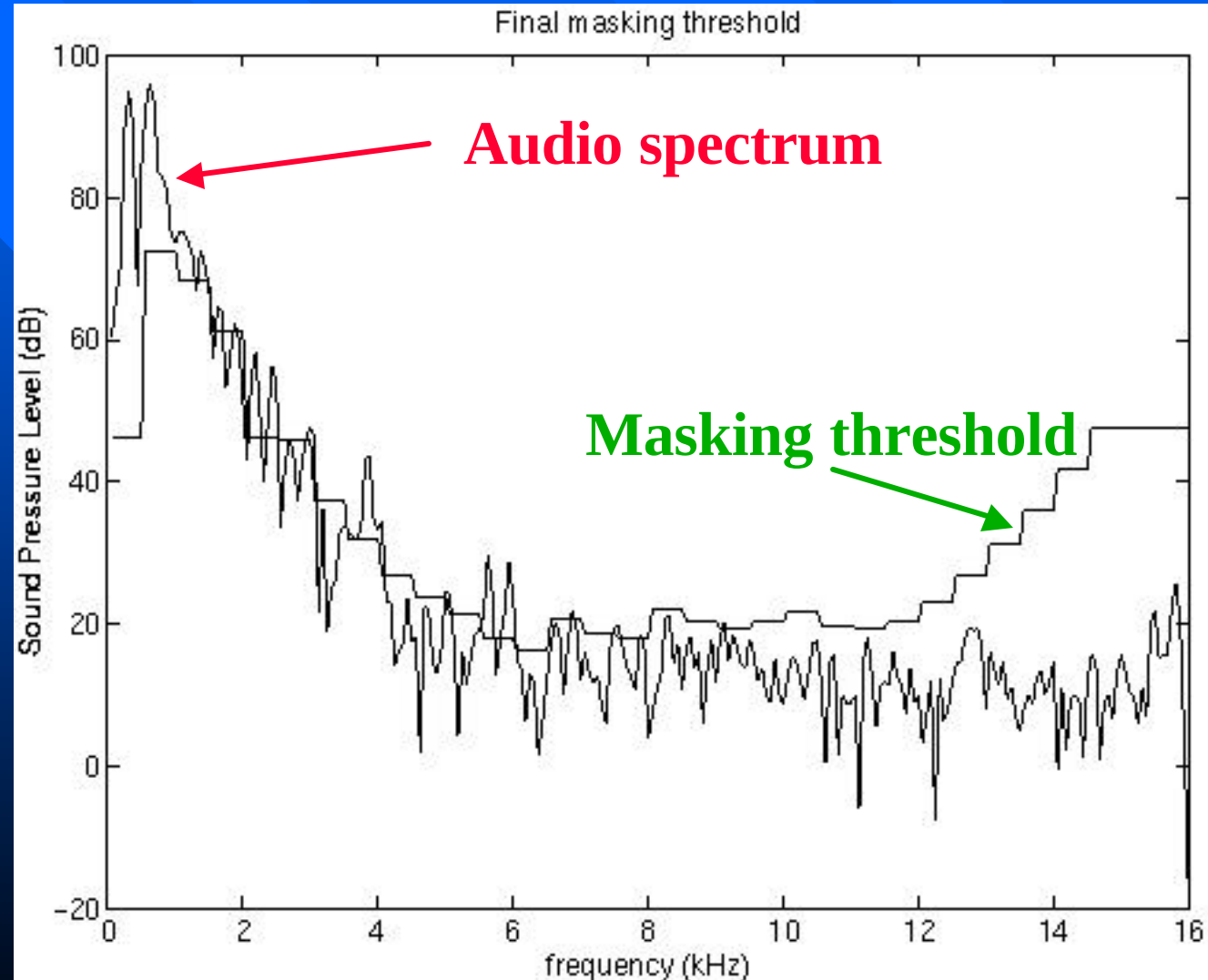
Transparency tested via blind tests

# Masking

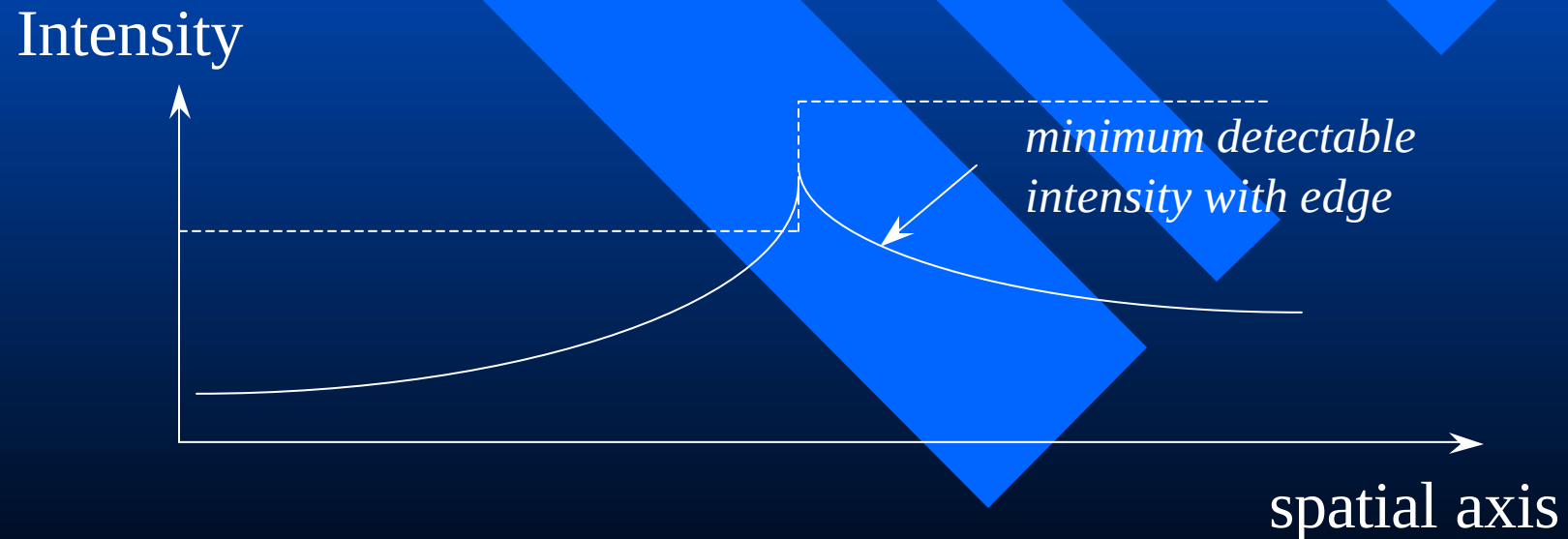
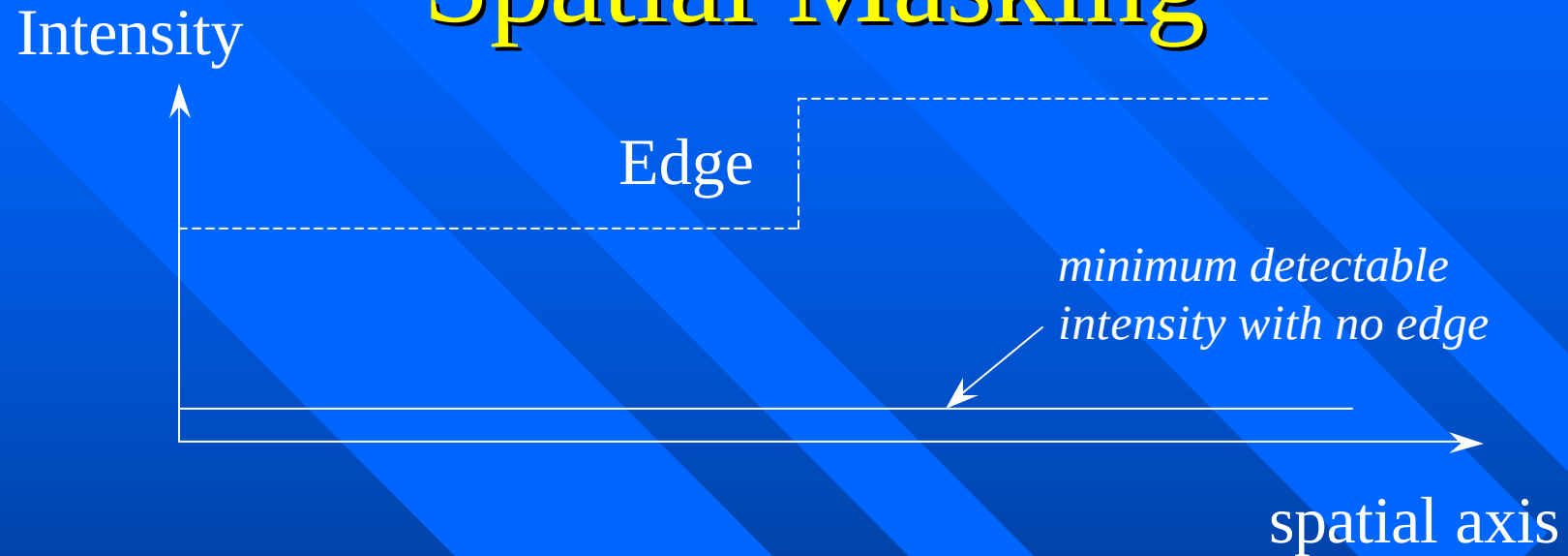
- Signal is perceptually inaudible or invisible in the presence of a masking signal
  - frequency masking
  - temporal masking
- Challenges:
  - variable embed rates
  - how to use masking at extract end?



# Example: Frequency masking in audio



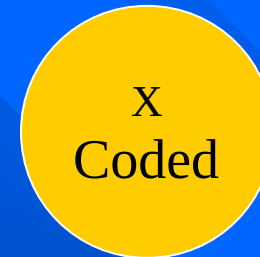
# Spatial Masking



# Data Embedding Issues

- Transparency
- Capacity
- Robustness
- Security

# Data Embedding Capacity



- Capacity depends on initial representation
- Capacity inversely proportional to robustness

| Host Signal | Byte rate or byte density                                            |
|-------------|----------------------------------------------------------------------|
| Audio       | <1 – 256 bytes/sec in mono CD quality audio signal                   |
| Image       | <1 – 256 bytes in 8 bits gray scale 128x128 image                    |
| Video       | 900 – 9216 bytes/sec in 320x240 8 bit gray scale 24 frames/sec video |

# Data Embedding Issues

- Transparency
- Capacity
- Robustness
- Security

# Robustness Issues

- Compression: JPEG, ZWT, etc.
- Filtering
- Additive noise
- Scaling and cropping
- Rotation
- Printing and Scanning

# SDMI Requirements

## Robust Watermark

- Transparency
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## Fragile Watermark

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- Anything but coding
- Low complexity

# Rotation and Scaling



*Original Image*



*Rotated two degrees  
and scaled*

# Cropping

- Only considers a small segment of the image

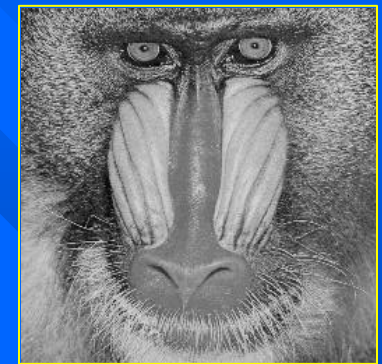


→ %15 cropping →

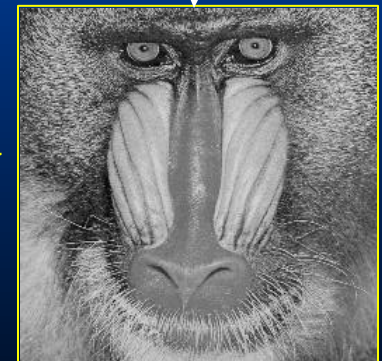


# Data Embedding Techniques

- Additive schemes
- Modulation (quantization) based schemes
- Self-Synchronizing schemes

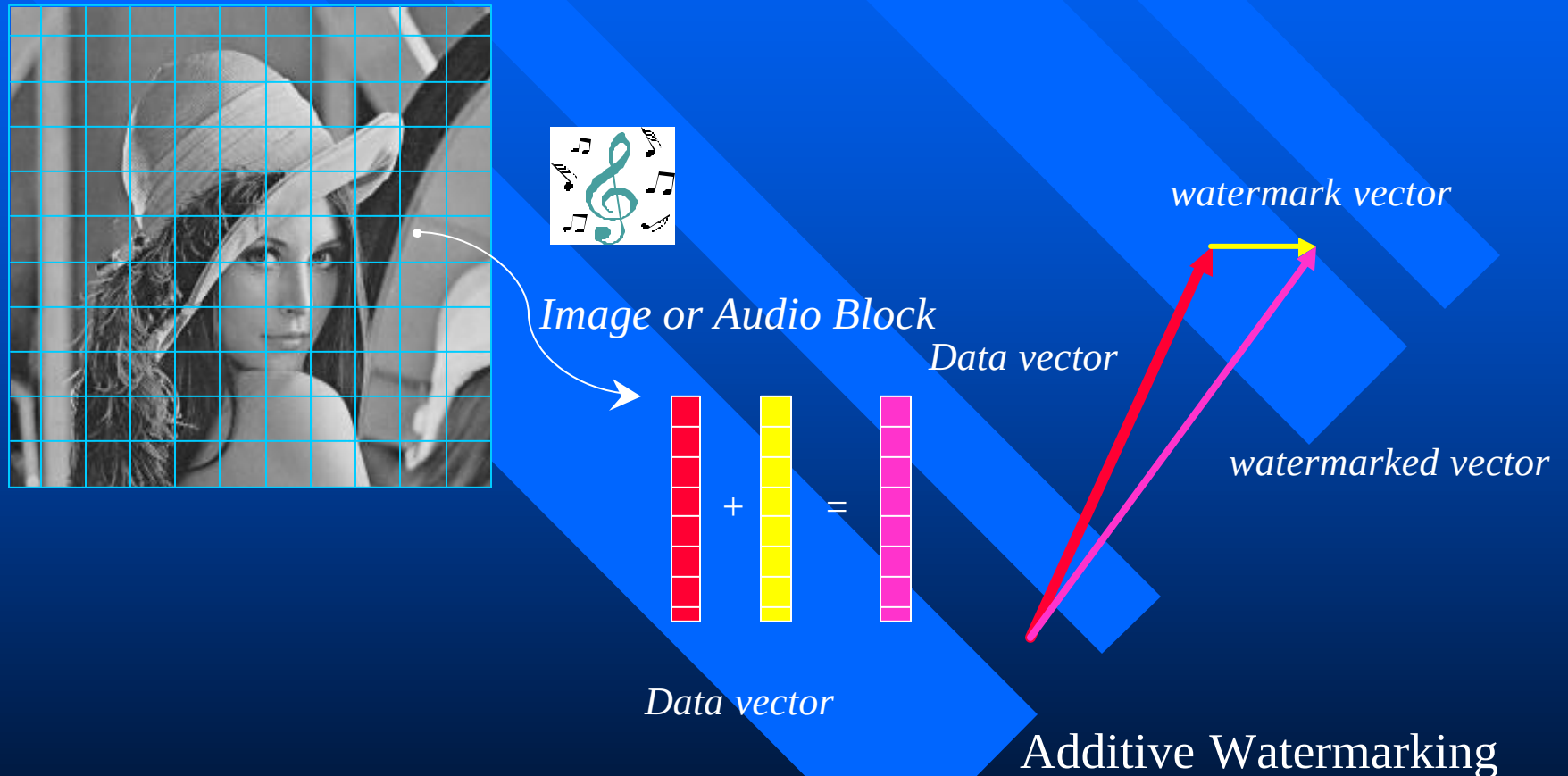


Embedded Info  
(e.g., text, audio,  
video, etc.)

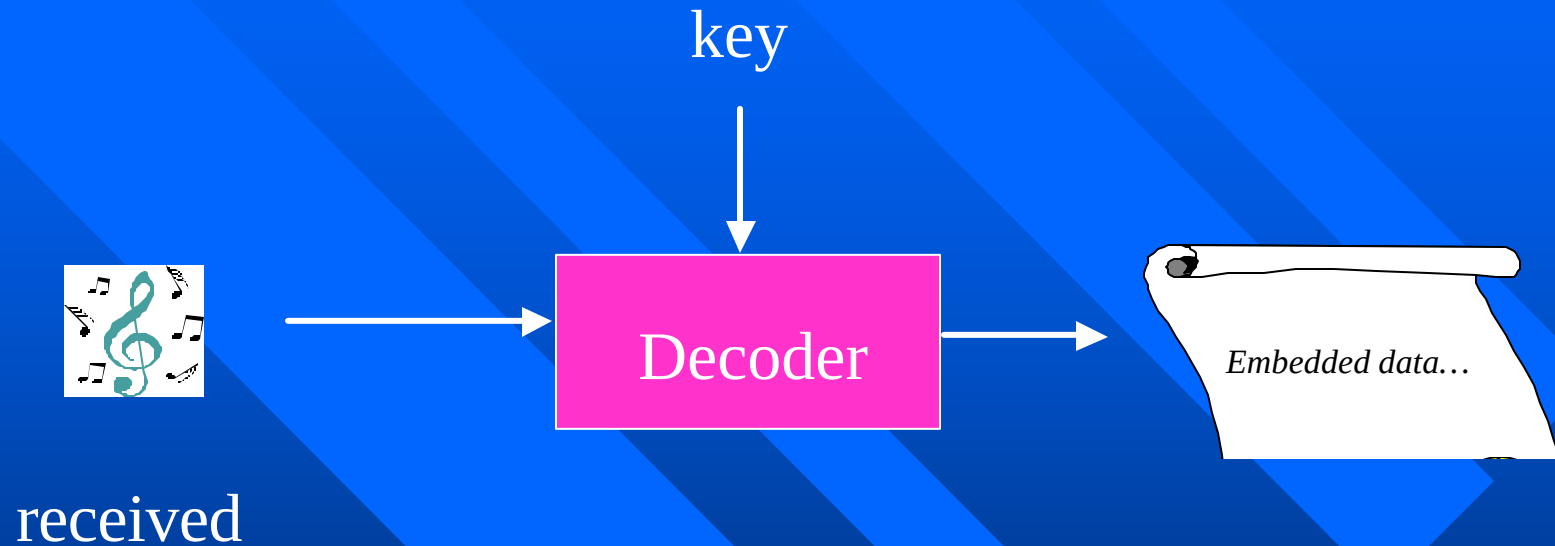


**Perceptually  
Unchanged**

# Additive Techniques



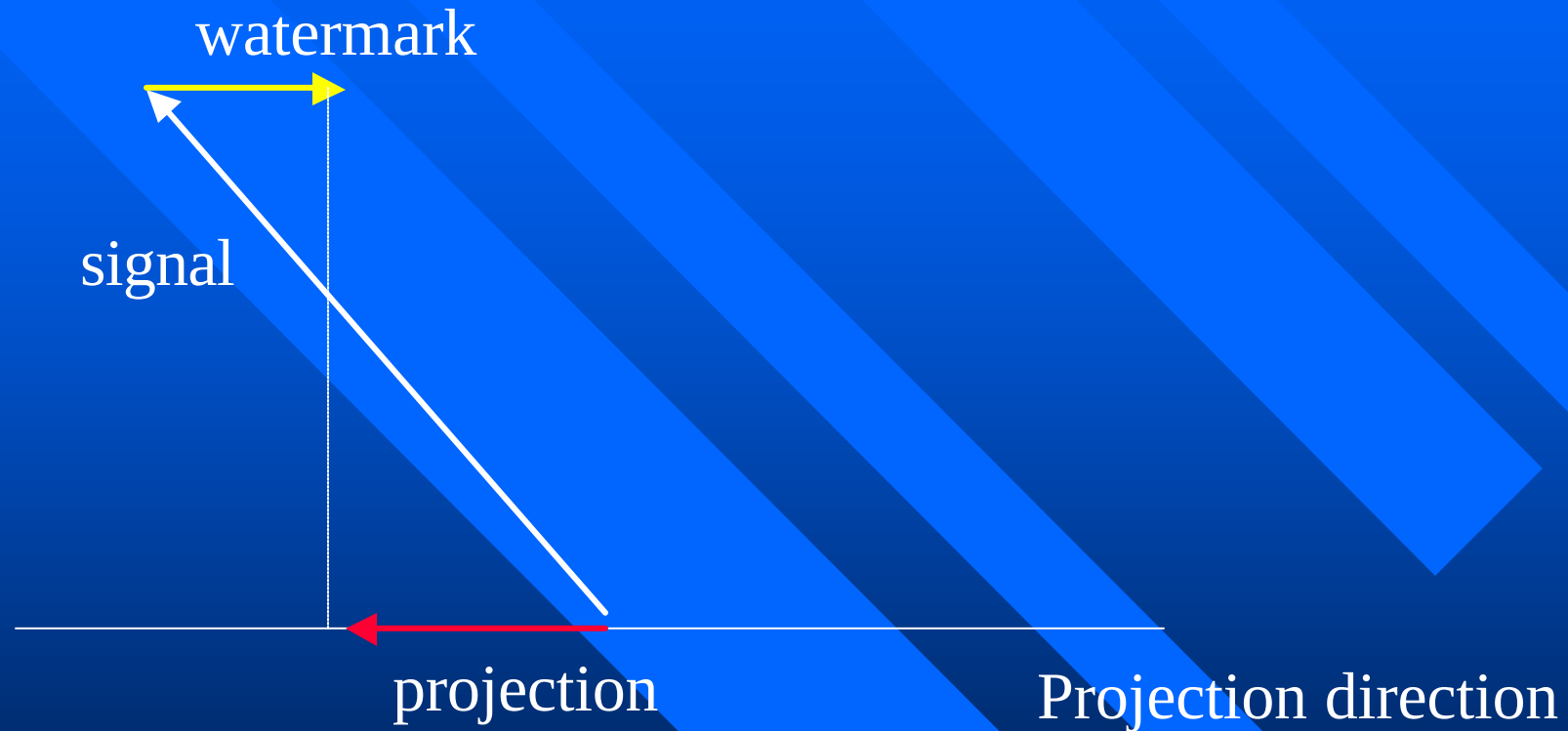
# Data Extraction



## Challenges:

- optimal use of masking
- dealing with host signal: major challenge for additive schemes
- dealing with distortions: synchronization problem

# Additive Techniques



- Strong signals confuse watermark detector.
- To decorrelate signal and watermark:
  - use properly designed watermarks
  - use longer blocks

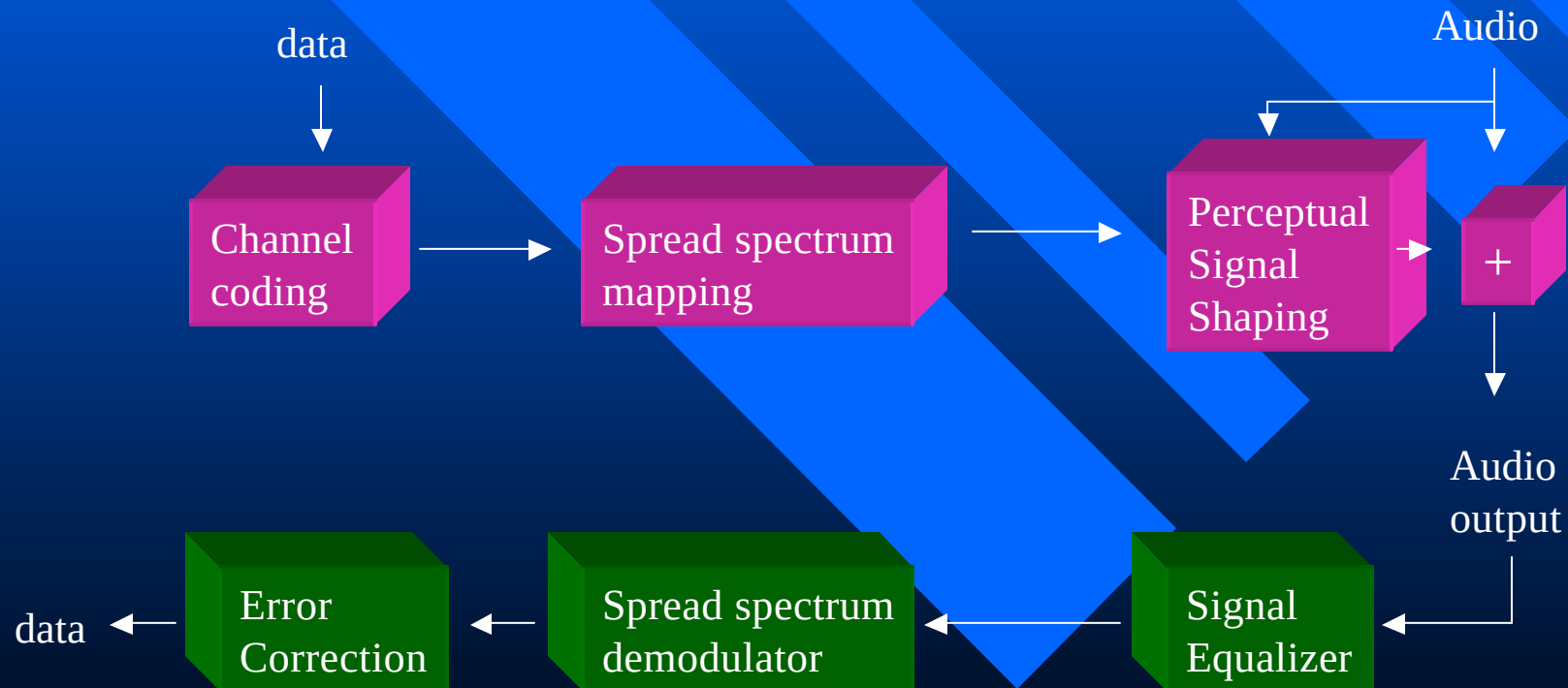
# Spread-spectrum Techniques

- Most popular approach
- Originated with BBN in audio domain
- Principle:  
*Add noise-like sequence in frequency domain or at random frequency, time or space locations to ensure that watermark and signal are uncorrelated.*

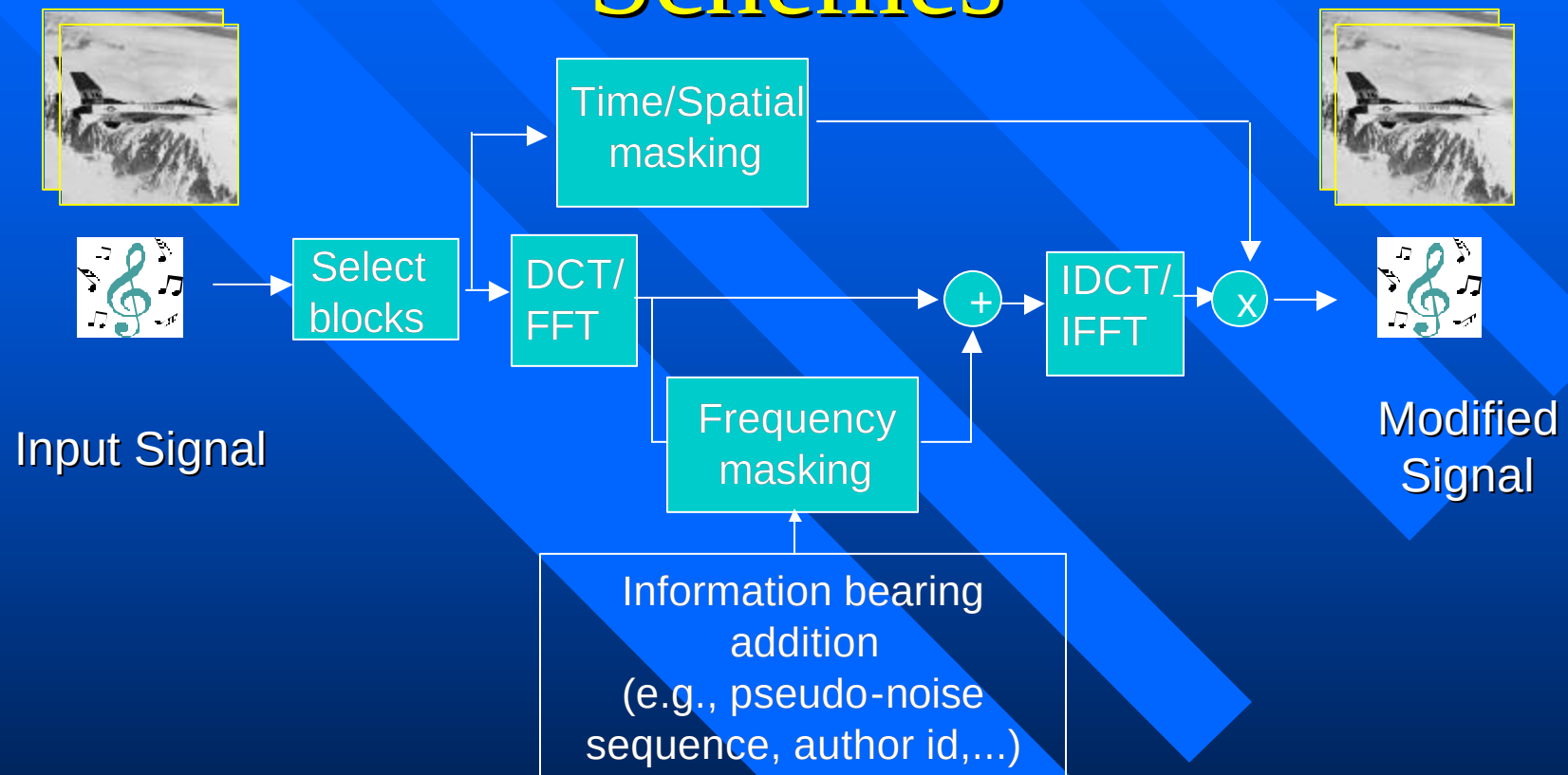


# Spread-spectrum Communications

## ■ Original BBN technique (1994 patent)



# Additive Data Embedding Schemes



- Spread-spectrum approaches
- Echo coding

# Spread-spectrum Communications

## ■ Advantages:

- secure, noise-like, undetectable
- robust to interference/attack

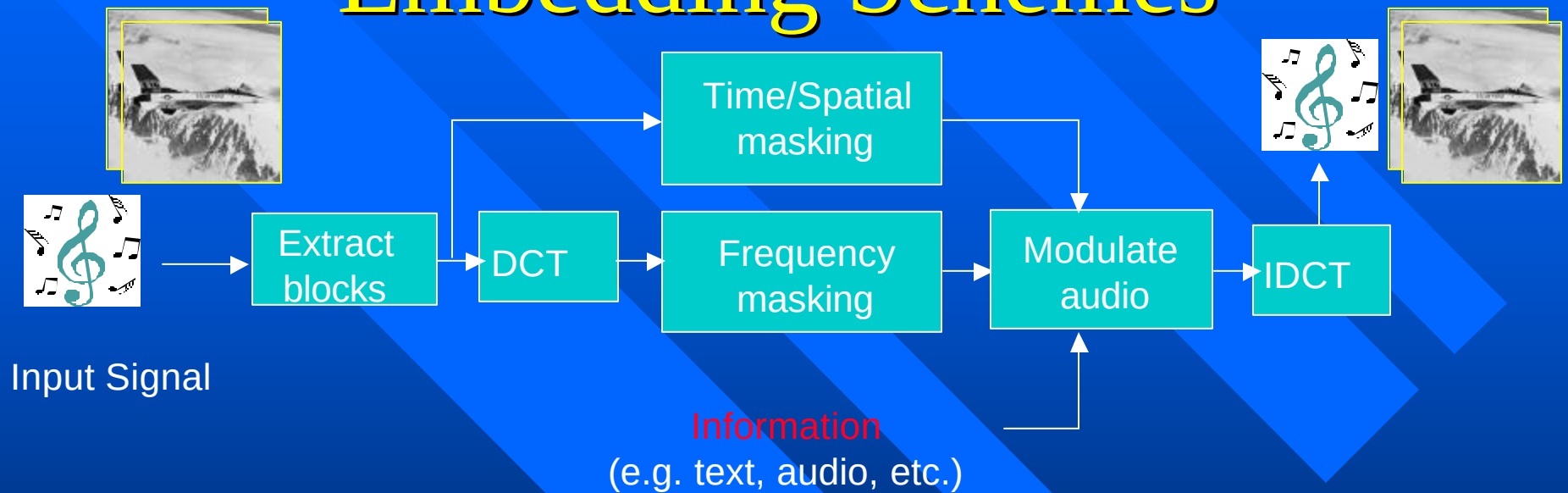


# Additive Techniques

## ■ Limitations:

- low capacity
- require proper synchronization
- masking challenge

# Modulation Based Data Embedding Schemes



- Least significant bit modification
- Phase modulation
- Band or patch replacement
- Quantized projections

# LSB Approaches

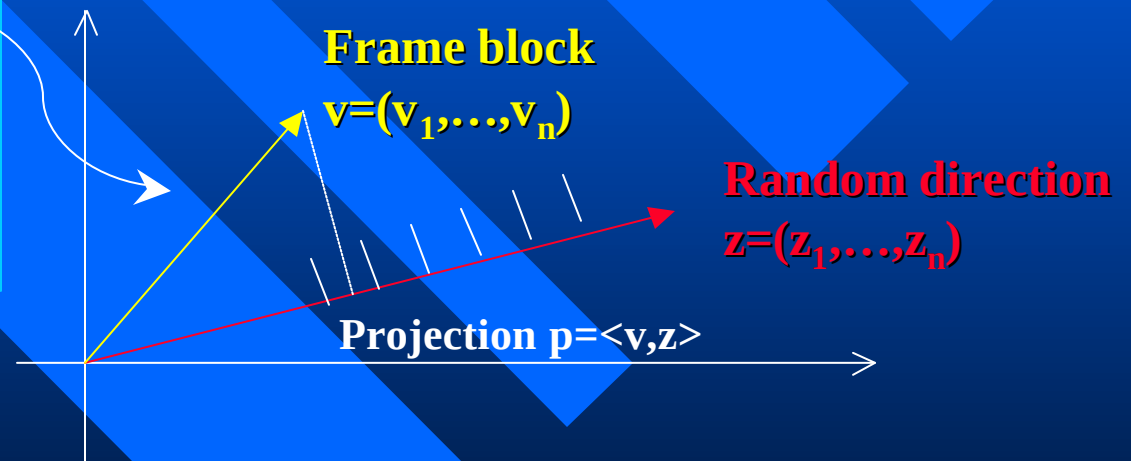
- Re-quantize coefficient values

$xxx\dots 1x=1$

$xxx\dots 0x=0$

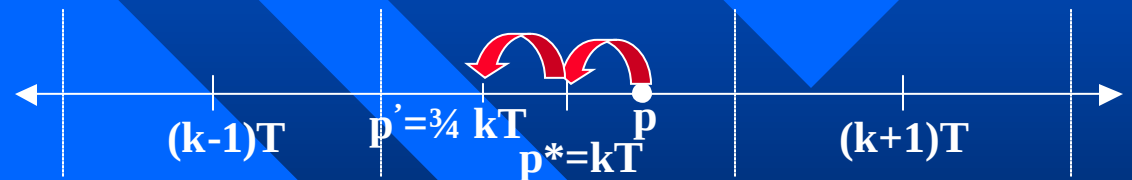
- Can embed 1 or more bits per sample with no distortion
- Requires error correction
- As more redundancy is added, tends to spread-spectrum approaches

# Projection Approach



# Quantization

Embedding data  
projection quantized by  
threshold  $T$   
Quantized projection  
perturbed by  $\pm T/4$



Embedding a '0'

$$\vec{v}' = \vec{v} + (p' - p) \cdot \vec{z}$$

# Projection Approach

## ■ Advantages

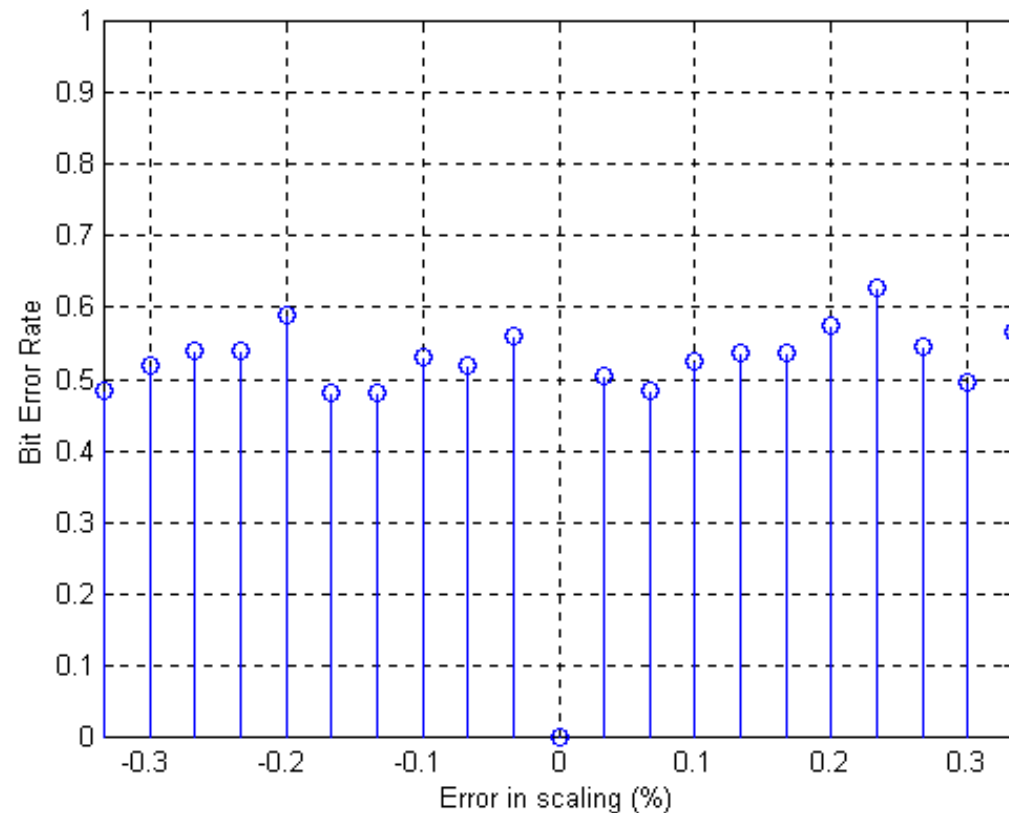
- large capacity
- ease of incorporating masking
- robustness

## ■ Limitations

- slow synchronization:
  - » sensitive to scaling and rotation,
  - » requires slow, exhaustive search

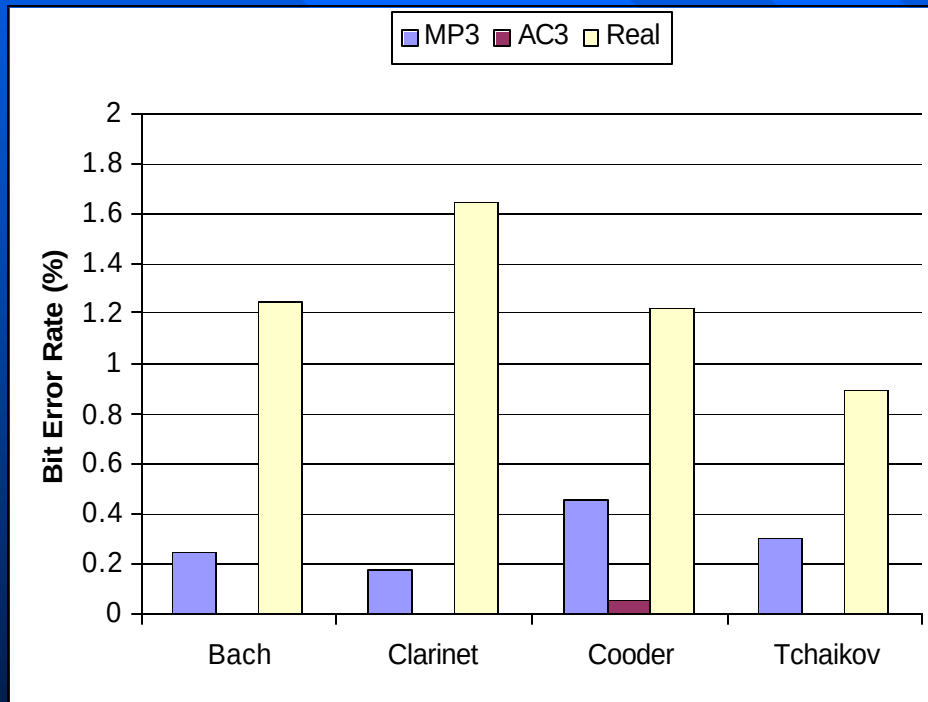
# Incorrect Scale

- Bit error rate at incorrect scale. 42 blocks of length 1024, 1 bit/block, DCT quantization



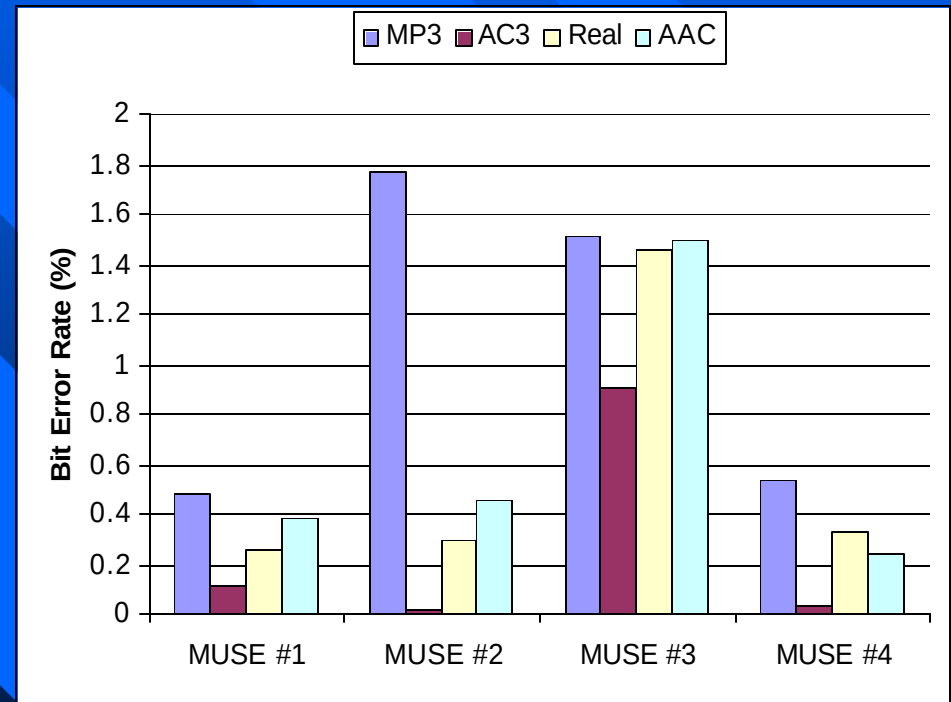
# Coding Robustness

■ Embedding rate: 42 bits/s. Random text.



## Mono

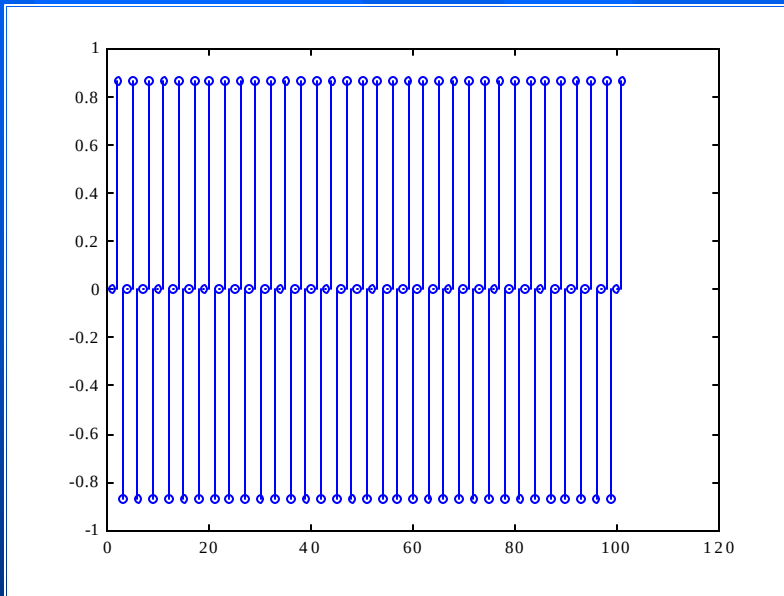
MP-3, AC-3: 56 Kbits/s  
Real: 40 Kbits/s (SR: 22.05 Ksamples/s)



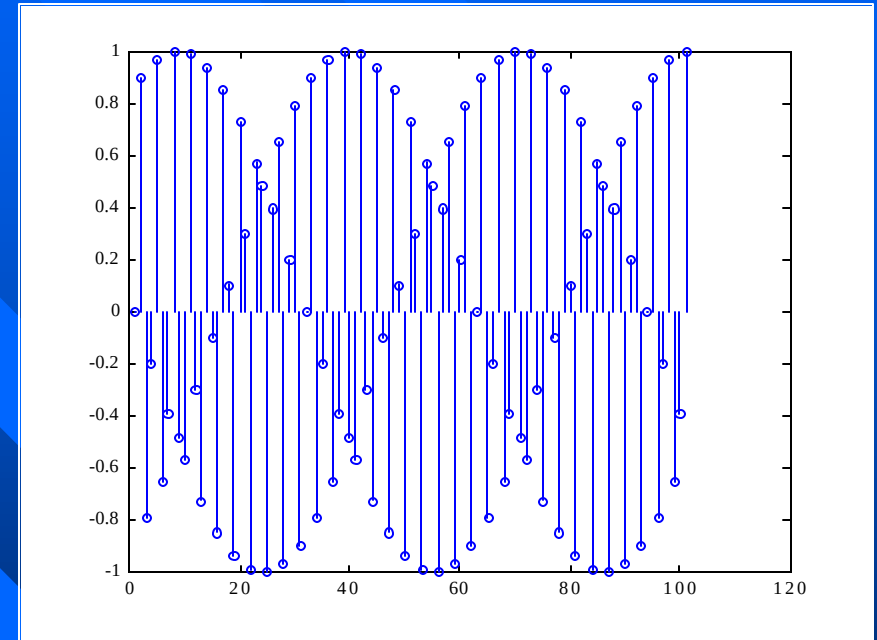
## Stereo

MP-3, AC-3, AAC: 96 Kbits/s  
Real: 80 Kbits/s (SR: 32 Ksamples/s)

# Synchronization Problems



50 Hz sinusoid  
sampled at 150 samples/s



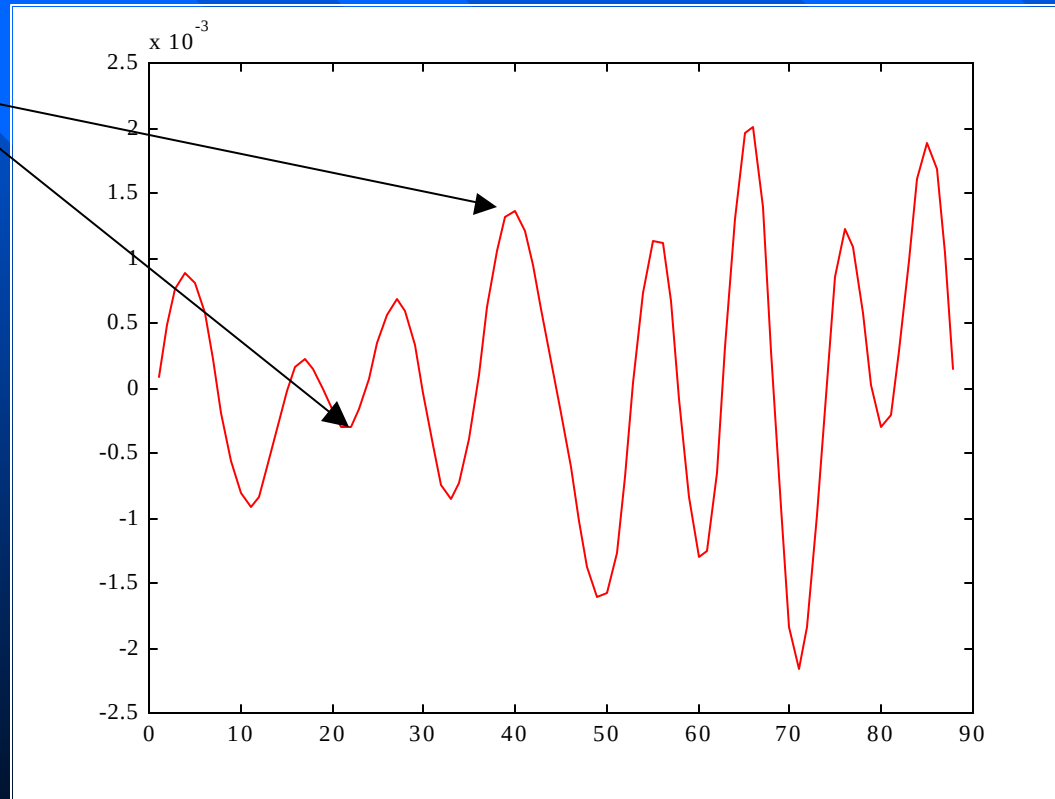
50 Hz sinusoid  
sampled at 155 samples/s

- Signal synchronization is needed for accurate extraction of the watermark

# Self-Synchronizing Approaches

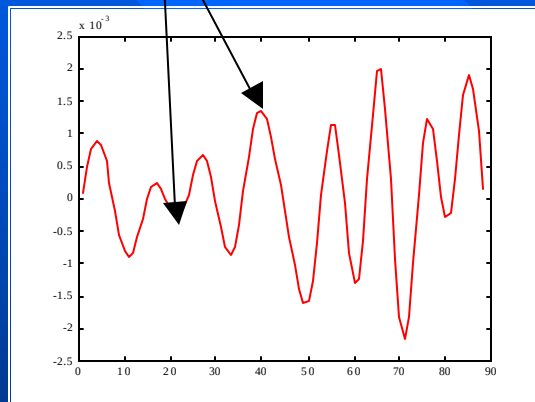
- Use adaptive “markers” to find the data

*Local extrema*



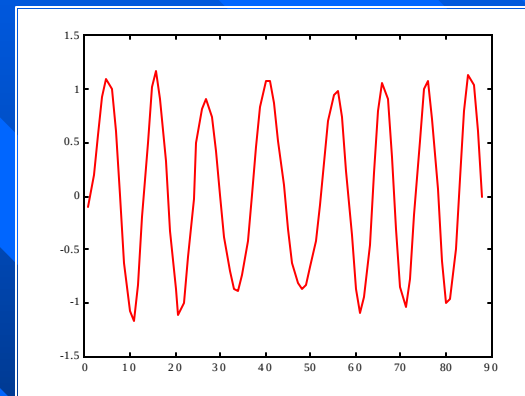
# Self-Synchronizing Approaches

*Local extrema*



original

+



Frequency and time  
domain constrained  
modification of extremas

Apply any of the known techniques to samples  
of the signal taken at extrema locations

# Comparisons

|                                                                  | Additive Techniques                                    | Modulation Techniques            | Self-Synchronizing Techniques     |
|------------------------------------------------------------------|--------------------------------------------------------|----------------------------------|-----------------------------------|
| <b>Masking</b>                                                   | Easily implemented                                     | Easily implemented               | Moderately difficult to implement |
| <b>Capacity</b>                                                  | Low                                                    | High                             | High                              |
| <b>Robustness</b><br>Coding<br>Time scale changes<br>Overdubbing | Moderately sensitive<br>Sensitive<br>Moderately robust | Robust<br>Sensitive<br>Sensitive | Robust<br>Robust<br>Sensitive     |
| <b>Realtime Extraction</b>                                       | Moderately fast                                        | Slow                             | Fast                              |

# Conclusion

- Watermarking is still in its infancy but has benefited tremendously from work on human perception
- Audio watermarking field is very active and solutions are being deployed in many applications
- Many challenges remain to meet user needs in some applications